

CHE 210 HW 5

P3.21.)

12 oz cup

200 mg caffeine

$$\frac{dm_{\text{sys}}}{dt} = -0.116 m_{\text{sys}} \quad \checkmark$$

 $m_{\text{sys}} [=] \text{ mg/hr}$ time for $m_{\text{sys}} \rightarrow 100 \text{ mg}$

- known info

at $t=0$, +200 mgat $t=8$, +200 mg

$$\frac{dm_c}{dt} = \text{rate in} - \text{rate out} = -0.116 m \quad \checkmark$$

$$\int \frac{1}{m} dm = \int -0.116 dt \rightarrow \ln(m) = -0.116t + C$$

 $\Rightarrow$ boundary condition: 200 mg at $t=0$ $\checkmark$

$$\ln(200) = -0.116(0) + C$$

$$\ln(200) = C = 5.30$$

$$\ln(100) = -0.116(t) + 5.30 \quad \checkmark$$

$$t = \frac{\ln(100) - 5.30}{-0.116} = 5.99 \text{ hr} \quad \checkmark$$

P3.21 cont.)

$$t = 5.99 \text{ hr}$$

$$\ln(m) = (8)(-0.116) + 5.30$$

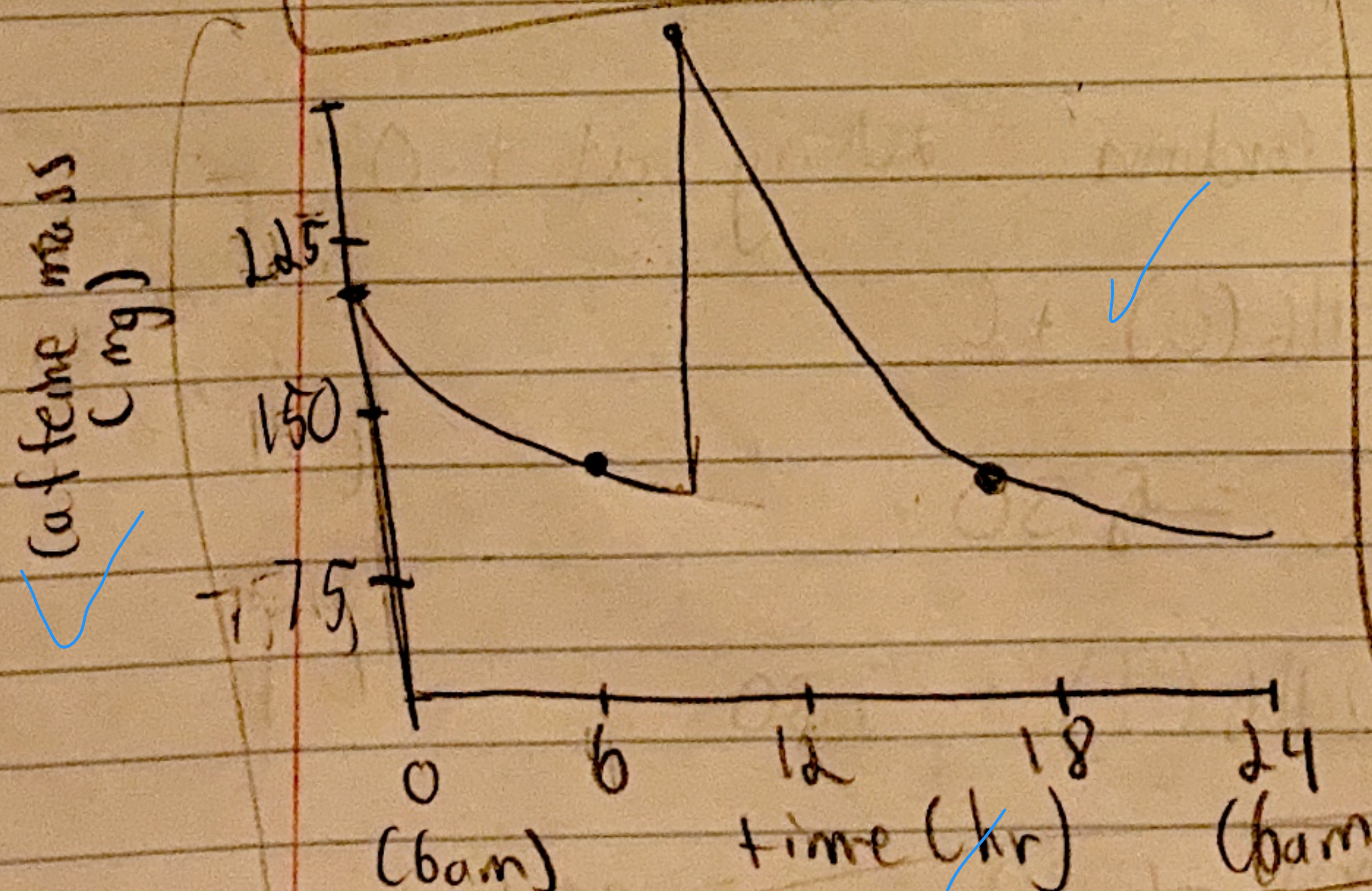
$$m = 79.20 \text{ mg}$$

$$\text{at 2 pm, } m = 79.20 + 200 = 279.2 \text{ mg}$$

$$\ln(279.2) = -0.116 + C$$
$$C = \ln(279.2) = 5.63$$

$$\ln(100) = -0.116t + 5.63$$
$$t = 8.83 \text{ hr}$$

After the first cup, it takes 5.99 hr to get back down to 100 mg in your body. After the second cup, it takes 8.83 hr.



$$\ln(m) = -0.116(9) + 5.63$$
$$m = 98.101$$

at 11pm, you still have nearly 100 mg of caffeine in you, so, yes, don't drink the 2pm coffee.

P3.29

rate of degradation = $2700(1 - e^{-0.3t})$ ✓
of immunotoxin

rate of immunotoxin leaving = 57000 molecules/hr

need at least 30000 molecules after 8 hr ✓

$$\frac{dm_{i, \text{accum}}}{dt} = m_{i, \text{in}} - m_{i, \text{leave}} - m_{i, \text{die}}$$

$$\frac{dm_{i, \text{accum}}}{dt} = 62000 - 57000 - (2700(1 - e^{-0.3t}))$$
 ✓

$$\int \frac{dm_{i, \text{a}}}{dt} = m_{i, \text{a}} = \int 2300 + 2700e^{-0.3t} = 1300t - 9000e^{-0.3t} + C$$
 ✓

$$0 = 1300(0) - 9000e^{-0.3(0)} + C$$
$$9000 = C$$

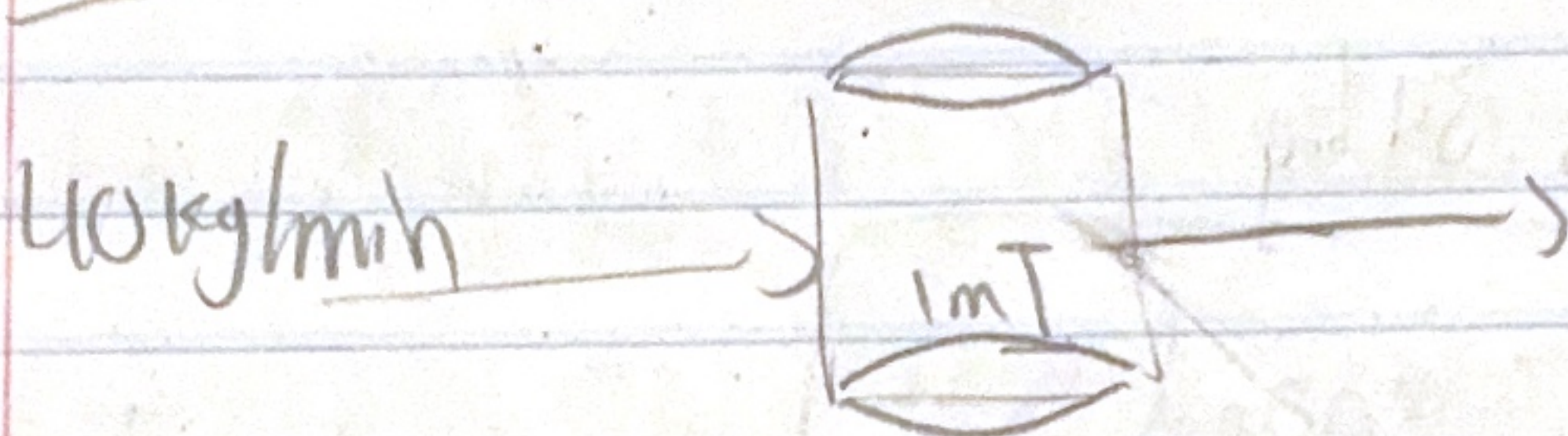
$$m_{i, \text{a}} = 1300t - 9000e^{-0.3t} + 9000$$
 ✓

$$m_{i, \text{a}}(8) = 1300(8) - 9000e^{-0.3(8)} + 9000 = 26583.53891$$
 ✓

after 8 hours the rate of accumulation is 26,583.54 molecules per hour.
This is too low, so do not invest. ✓

15/15

P3.41



$$\rho_x = 1000 \text{ kg/m}^3$$

$$D = 1 \text{ m}, H = 3 \text{ m}$$

$$\dot{m}_{\text{leak}} = 4 \cdot (\text{liq. height} - \text{tank height at leak})$$

at ~~40~~ at $t = 40 \text{ min}$, how much leaked out? 1m

$$A_{\text{area}} = 3.14 \cdot (0.5)^2 = 0.7854 \text{ m}^2$$

$$\frac{dh}{dt} = \frac{0.04}{0.7854} = 0.05093 \text{ m/min}$$

change in height

$$t = \frac{1 \text{ m}}{0.05093} = 19.64 \text{ min}$$

$$\dot{m} = 4\sqrt{h-1} \quad v = \dot{m}/\rho = 0.004\sqrt{h-1}$$

$$\frac{dh}{dt} = \frac{0.04 - 0.004\sqrt{h-1}}{A} = h(19.64) = 1$$

$$dt = \frac{2A}{0.04 - 0.004\sqrt{h-1}} dy$$

$$M_c = \int_0^{40} \dot{m} dt = \int_0^{40} 4\sqrt{h-1} \frac{2A}{0.04 - 0.004\sqrt{h-1}} dy$$

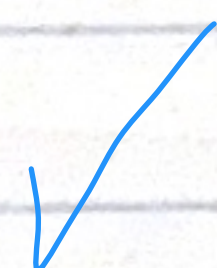
$$= 8A \int_0^{40} \frac{\sqrt{h-1}}{0.04 - 0.004\sqrt{h-1}} dy$$

$$-115.4\sqrt{h-1} - 2500\sqrt{h-1} - 25000 \ln(0.04 - 0.004\sqrt{h-1})$$

$$8(0.7854)(-129y^2 - 2500y - 2500\ln(10-y)) + 2500\ln(10) = 539$$

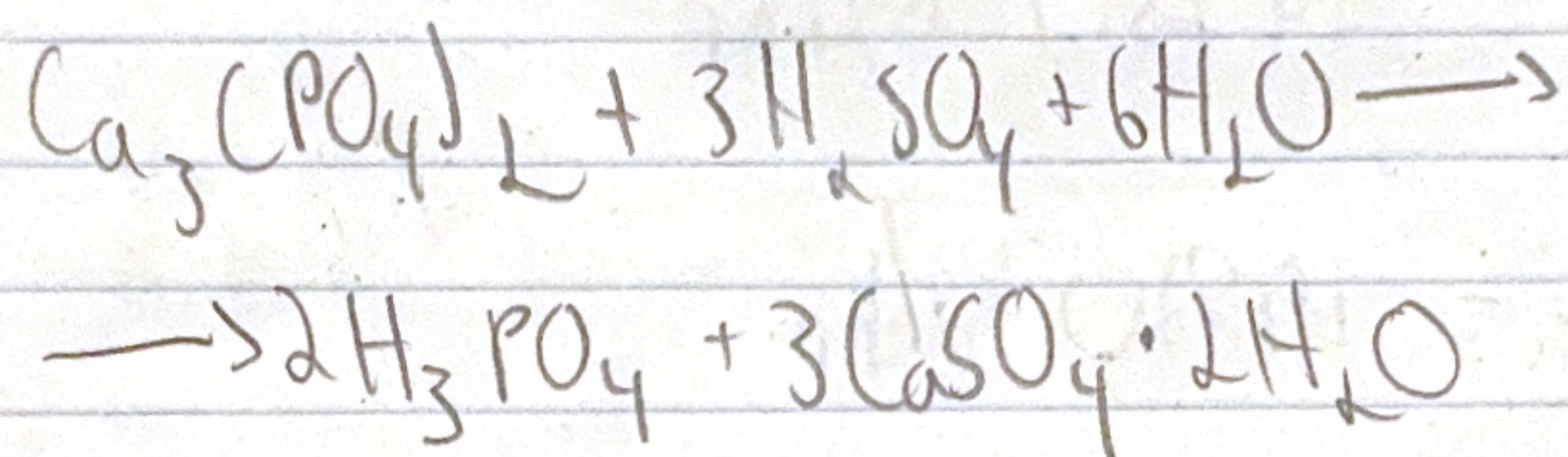
$$0.1 \cdot 53.9 = 5.39 \text{ kg}$$

5.39 kg of compound X

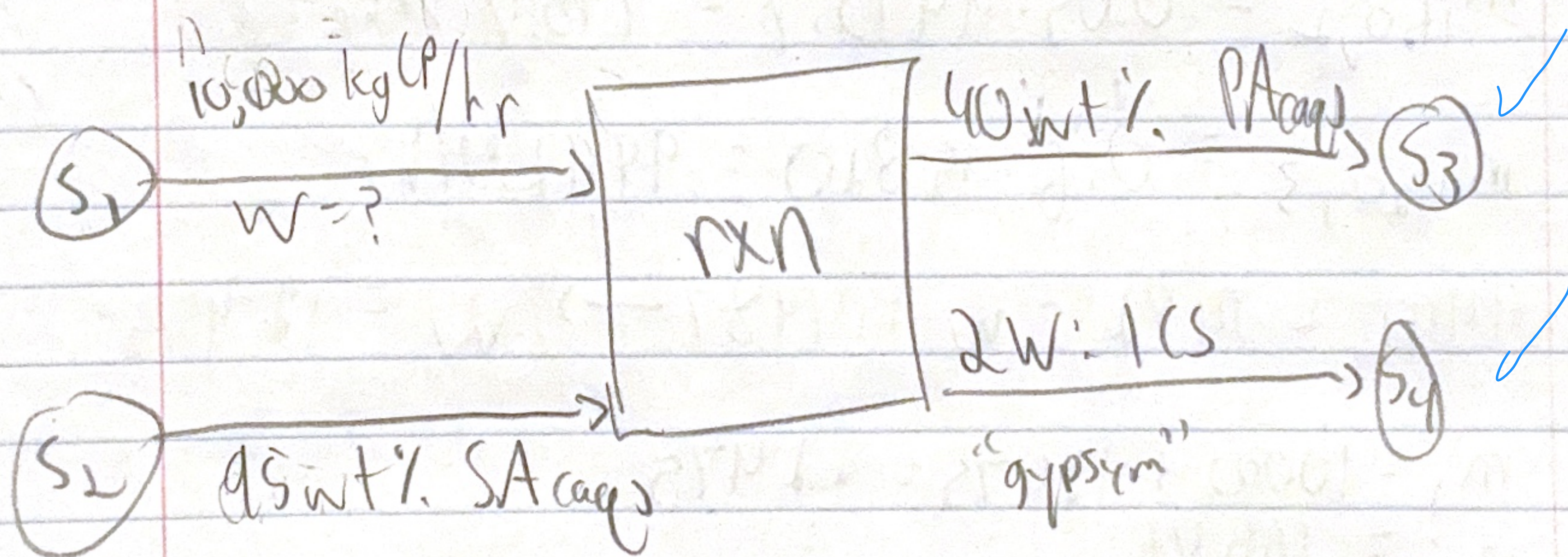


20/20

Problem 4.



✓
✓ good



$$n_{\text{CP}} = 10,000 / 310.18 = 32.233 \text{ kmol/hr}$$

↑
molar mass CP

$$n_{\text{H}_2\text{SO}_4} = 96.699 \text{ kmol/hr}$$

$$n_{\text{H}_3\text{PO}_4} = 64.466$$

$$n_{\text{CaSO}_4 \cdot 2\text{H}_2\text{O}} = 96.699$$

$$n_{\text{H}_2\text{O}} = 193.398$$

→ convert to kg
via molar masses

$$m_{\text{H}_2\text{SO}_4} = 9475 \text{ kg/hr}$$

$$m_{\text{H}_3\text{PO}_4} = 6328$$

$$m_{\text{CaSO}_4 \cdot 2\text{H}_2\text{O}} = 16646$$

$$m_{\text{H}_2\text{O}} = 3482$$

$$\dot{m}_2 = \frac{9475}{0.95} = 9973.7 \text{ kg/hr} \quad \checkmark$$

$$\dot{m}_3 = \frac{6328}{0.4} = 15820 \text{ kg/hr} \quad \checkmark$$

$$\dot{m}_{H_2O,2} = 0.05 \cdot 9973.7 = 498.7 \text{ kg/hr}$$

$$\dot{m}_{H_2O,3} = 0.6 \cdot 15820 = 9492 \text{ kg/hr} \quad \checkmark$$

$$9492 + 3482 = w + 4987 \rightarrow w = 12475$$

$$\dot{m}_1 = 10000 + 12475 = 22475$$

$$\dot{m}_4 = 16646$$

a.)

$$\dot{m}_2 = 9973.7$$

$$\dot{m}_3 = 15820$$

$$\dot{m}_{w,2} = 498.7$$

$$\dot{m}_{w,3} = 9492$$

$$\dot{m}_{w,1} = 12475$$

$$\dot{m}_1 = 22475$$

$$\dot{m}_4 = 16646$$

all kg/hr

20
20

10000 kg at 20 wt% H_3PO_4
2000 kg acid
8000 kg H_2O

$$0.4 \cdot 15820 = 6328 \text{ kg}$$

$$2000 + 6328 = 8328 \text{ kg}$$

$$10000 + 15820 = 25820 \text{ kg}$$

b.) 32.25 wt%

$$\rightarrow \frac{8328}{25820} \cdot 100\% = 32.25\% \quad \checkmark$$

Problem 5c)

$$V_0 = 1000 \text{ L}$$

95 wt% EtOH

$$Q = 100 \text{ L/min}$$

$$\frac{d(Mw)}{dt} = -\dot{m}_{\text{out}} w, \quad \dot{m}_{\text{out}} = \dot{m}_{\text{in}} = P_w Q$$

$$\frac{dw}{dt} = -\frac{P_w Q}{M} w$$

$$w(t) = w_0 e^{-kt}$$

$$k = \frac{P_w Q}{M}$$

$$t = (\ln 0.95 - \ln 0.05) \cdot \frac{1}{k} = \frac{M}{P_w Q} \ln(19)$$

$$t = \frac{0.806 \cdot 10^3}{0.998 \cdot 10^2} \cdot \ln 19 = 23.8$$

$$\boxed{a.) 23.8 \text{ min}}$$

$$\frac{d}{dt} w p(w) V = -w p(w) Q$$

↓ product rate

$$V(p(w) + w p'(w)) \frac{dw}{dt} = -w p(w) Q$$

$$p(w) = 0.998 - 0.201 w$$

$$\int_{0.95}^{0.05} \frac{p(w) + w p'(w)}{w p(w)} dw = -\frac{Q}{V} t$$

$$t = \frac{V}{Q} \ln \left(\frac{W_0(a - cW_0)}{W_f(a - cW_f)} \right)$$

a is 0.998

c is 0.209

$$\frac{V}{Q} = 10 \text{ min}$$

$$t = 10 \cdot \ln \left(\frac{0.95(0.998 - [0.209 \cdot 0.95])}{0.05(0.998 - [0.209 \cdot 0.05])} \right)$$

$$= 27.3$$

b.) 27.3 minutes

$\frac{30}{30}$